

hw2

Ian Chi

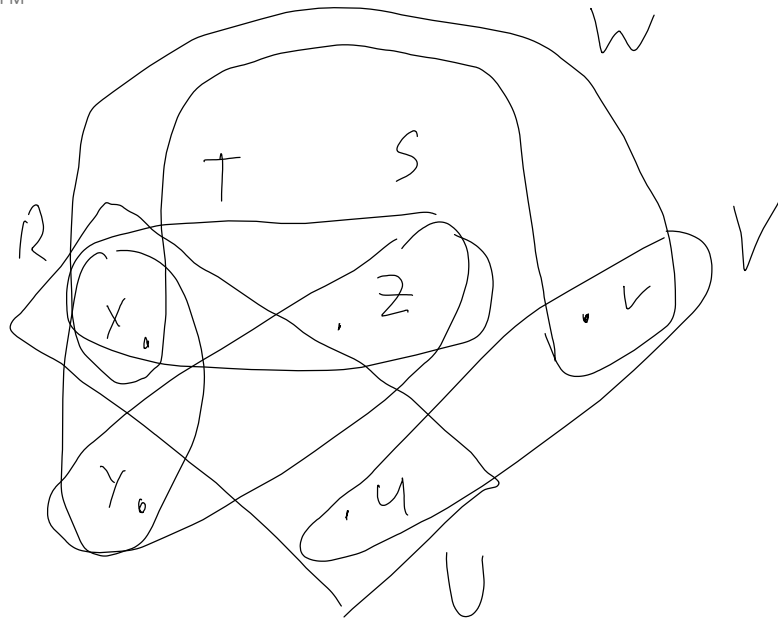
March 8, 2026

1

Hw3

Sunday, March 8, 2026 4:30 PM

a)



1) 2.5

R	S	T	U	V	W
0	1	0	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$

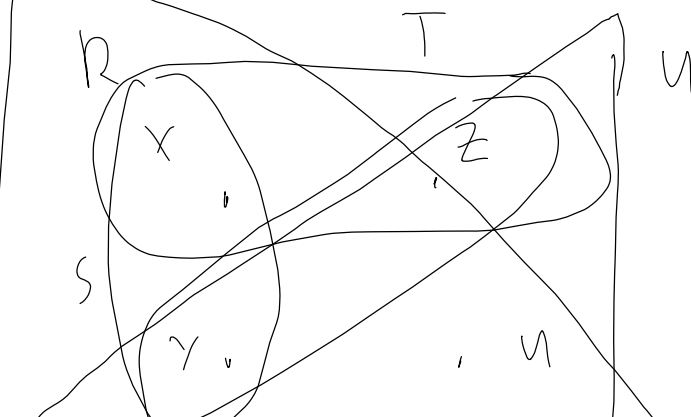
2) $W(\text{count}) = n^{2.5}$

3) 2.5

X	Y	Z	U	V
$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$

4) $n^{2.5}$

b)





1) 1.5

R	S	T	U	V
0	0	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$

2) $h^{1.5}$

3) 1.5

X	Y	Z	U
$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	0

4) $n^{1.5}$

2

1. the width does not depend on k . the width remains 2 for all $k \geq 3$

```
2. SELECT COUNT(*)
   FROM edges e1
   JOIN edges e2 ON e1.dst = e2.src
   JOIN edges e3 ON e2.dst = e3.src
   JOIN edges e4 ON e3.dst = e4.src
   WHERE e4.dst = e1.src;
```

178704122 this took 42437.452 ms (00:42.437)

```
3.      WITH
p23 AS (
    SELECT e1.src AS x2, e1.dst AS x3, e2.dst AS x4
    FROM edges e1
    JOIN edges e2 ON e1.dst = e2.src
),

p1234 AS (
    SELECT e.src AS x1, p.x2, p.x3, p.x4
    FROM edges e
    JOIN p23 p ON e.dst = p.x2
),

cycles AS (
    SELECT *
    FROM p1234 p
    JOIN edges e ON p.x4 = e.src AND e.dst = p.x1
)

SELECT COUNT(*) FROM cycles;
```

178704122 Time: 44329.151 ms (00:44.329)

it took almost the same amount of time, slightly longer. it has some overhead
i guess?